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*计算题.

2.1 (1) $a=[2,-1,3]^T$, $b=[1,4,-2]^T$

$a\cdot b = 2\times1+(-1)\times4+3\times(-2) = -8$

(2) $A=\begin{bmatrix}1&0&2\\-1&3&1\end{bmatrix}_{2\times3}$ $B=\begin{bmatrix}2&1\\0&-1\\3&2\end{bmatrix}_{3\times2}$

$C=A\times B=\begin{bmatrix}8&5\\1&-2\end{bmatrix}_{2\times2}$

(3) $|a|_F=\sqrt{2^2+(-1)^2+3^2}=\sqrt{14}$

3.1 解：设事件D为患病，T^+表示检测阳性

$P(D)=0.1\%$, $P(T^+|D)=0.99$, $P(T^+|\bar{D})=0.02$

$$P(D|T^+)=\frac{P(T^+|D)P(D)}{P(T^+|D)\cdot P(D)+P(T^+|\bar{D})P(\bar{D})}$$

$$=\frac{0.99\times0.001}{0.99\times0.001+0.02\times0.999}\approx0.0472.$$

故真正患病的概率为4.72%

4.1 解 $Z=(w_1x_1+w_2x_2-y)^2$, $x_1=2$, $x_2=1$, $y=3$.

(1) 令 $u=w_1x_1+w_2x_2-y$, $z=u^2$.

$$\frac{\partial z}{\partial w_1}=2u\cdot\frac{\partial u}{\partial w_1}=2(w_1x_1+w_2x_2-y)\cdot x_1$$

$$\frac{\partial z}{\partial w_2}=2u\cdot\frac{\partial u}{\partial w_2}=2(w_1x_1+w_2x_2-y)\cdot x_2$$

代入得：$\frac{\partial z}{\partial w_1}=4(2w_1+w_2-3)$

$$\frac{\partial z}{\partial w_2}=2(2w_1+w_2-3)$$

(2) $W_1=0.5$, $W_2=1$ 时的梯度值.

$$2W_1+W_2-3=2\times0.5+1-3=-1$$

$$\frac{\partial z}{\partial W_1}=4\times(-1)=-4.$$

$$\frac{\partial z}{\partial W_2}=-2.$$

∴ 梯度为 $(-4,-2)$

5.1 $y=wx+b$.

均方误差 $L=\frac{1}{n}\sum_{i=1}^{n}(y_i-(wx_i+b))^2$.

对 w 求偏导:

$$\frac{\partial L}{\partial w}=\frac{1}{n}\sum_{i=1}^{n}2(y_i-wx_i-b)(-x_i)$$

$$=-\frac{2}{n}\sum_{i=1}^{n}x_i(y_i-wx_i-b)$$

对 b 求偏导:

$$\frac{\partial L}{\partial b}=\frac{1}{n}\sum_{i=1}^{n}2(y_i-wx_i-b)(-1)$$

$$=-\frac{2}{n}\sum_{i=1}^{n}(y_i-wx_i-b)$$

6.1. 样本 $x_1, x_2, \cdots, x_n \sim N(\mu, \sigma^2)$ 独立同分布

(1) 似然函数:

每个样本的概率密度函数为:

$$f(x_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)$$

∵ 样本独立、似然函数为各密度乘积

$$L(\mu, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)$$

$$= (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right)$$

(2) $\ln L(\mu, \sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2.$

~~令~~ $\frac{\partial \ln L(\mu, \sigma^2)}{\partial \mu} = -\frac{1}{2\sigma^2} \cdot \frac{\partial}{\partial \mu}\sum_{i=1}^{n}(x_i-\mu)^2$

$\frac{\partial}{\partial \mu}(x_i-\mu)^2 = -2(x_i-\mu)$ ∴ $\frac{\partial \ln L}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu)$

~~令其为0.~~ 令其为0 ⇒ $\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$

∴ $\mu = \bar{x}$

(3) $\theta = \sigma^2$.

∴ $\ln L(\theta) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln\theta - \frac{1}{2\theta}\sum_{i=1}^{n}(x_i-\hat{\mu})^2$

令 $S = \sum_{i=1}^{n}(x_i-\hat{\mu})^2$. ⇒ $\ln L(\theta) = -\frac{n}{2}\ln\theta - \frac{S}{2\theta} + C$

$\frac{\partial \ln L(\theta)}{\partial \theta} = -\frac{n}{2\theta} + \frac{S}{2\theta^2} = 0$

$\theta = \frac{S}{n}$ ∴ $\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i-\hat{\mu})^2.$